

Structural Calculation Report

RC Beam Design to BS EN 1992-1-1:2004

EXECUTIVE SUMMARY

Overall Status:	FAIL	Max Utilisation:	123.5%
Section:	300.0x500.0 mm	Main Steel:	3-Ø16

Recommendation: Provided steel insufficient. Use 1-D32 ($A_s=804.2 \text{ mm}^2$, util.=80%).

1. Design Parameters

Concrete Grade	C30.0 ($f_{ck} = 30.0 \text{ MPa}$)
Steel Strength	$f_{yk} = 500.0 \text{ MPa}$
Nominal Cover	30.0 mm
Design Moment (M _{Ed})	120.0 kNm
Design Shear (V _{Ed})	80.0 kN

2. Bending Design

Effective depth:

$$d = h - c_{\text{nom}} - d_{\text{link}} - d_{\text{bar}}/2$$

$$d = 500.0 - 30.0 - 8.0 - 16.0/2$$

-> **454.0 mm**

Design concrete strength:

$$f_{cd} = \alpha_{cc} * f_{ck} / \gamma_c$$

$$f_{cd} = 0.85 * 30.0 / 1.5$$

-> **17.00 MPa**

Design steel strength:

$$f_{yd} = f_{yk} / \gamma_s$$

$$f_{yd} = 500.0 / 1.15$$

-> **434.78 MPa**

K-factor:

$$K = M_{Ed} / (b_{\text{eff}} * d^2 * f_{ck})$$

$$K = 120.0 * 10^6 / (300.0 * 454.0^2 * 30.0)$$

-> **0.0647 ($K' = 0.167$)**

Lever arm:

$$z = d * (0.5 + \sqrt{0.25 - K/1.134})$$

$$z = 454.0 * (0.5 + \sqrt{0.25 - 0.0647/1.134})$$

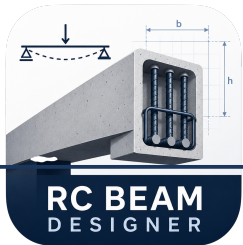
-> **426.43 mm**

Required tension steel:

$$A_{s,\text{req}} = M_{Ed} / (f_{yd} * z)$$

$$A_{s,\text{req}} = 120.0 * 10^6 / (434.78 * 426.43)$$

-> **647.24 mm²**



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Provided tension steel:

$$A_{s,prov} = n \cdot \pi \cdot d^2 / 4$$

$$A_{s,prov} = 3 \cdot \pi \cdot 16.0^2 / 4$$

-> **603.19 mm²**

3. Shear Design

Size factor:

$$k = \min(2.0, 1 + \sqrt{200/d})$$

$$k = 1 + \sqrt{200/454.0}$$

-> **1.664**

Concrete shear resistance:

$$V_{Rd,c} = [C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3}] \cdot b_w \cdot d$$

$$\rho_l = 0.00443$$

-> **64.40 kN**

Max strut capacity:

$$V_{Rd,max} = \alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd} / (\cot \theta + \tan \theta)$$

$$\theta = 21.8 \text{ deg}, \cot \theta = 2.50$$

-> **446.36 kN**

Required shear reinforcement:

$$A_{sw/s} = V_{Ed} / (z \cdot f_{ywd} \cdot \cot \theta)$$

$$A_{sw/s} = 80.0 \cdot 10^3 / (408.6 \cdot 434.8 \cdot 2.50)$$

-> **0.263 mm²/mm**

4. Deflection Check

Reference reinforcement ratio:

$$\rho_0 = \sqrt{f_{ck}} \cdot 10^{-3}$$

$$\rho_0 = \sqrt{30.0} \cdot 10^{-3}$$

-> **0.00548**

Actual reinforcement ratio:

$$\rho = A_{s,req} / (b \cdot d)$$

$$\rho = 647.24 / (300.0 \cdot 454.0)$$

-> **0.00475**

Basic span/depth ratio:

$$L/d_{basic} = K \cdot [11 + 1.5 \cdot \sqrt{f_{ck}} \cdot \rho_0 / \rho + \dots]$$

$$K_{sys} = 1.0$$

-> **21.51**

Modification factor:

$$\text{Factor} = (500/f_{yk}) \cdot (A_{s,prov}/A_{s,req}) \leq 1.5$$

$$\text{Factor} = (500/500.0) \cdot (603.2/647.2)$$

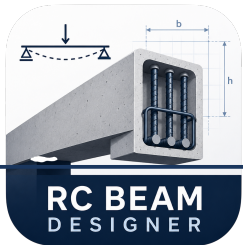
-> **0.932**

Allowable span/depth:

$$L/d_{allow} = L/d_{basic} \cdot \text{Factor}$$

$$21.51 \cdot 0.932$$

-> **20.05**



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Actual span/depth:

$$L/d_{\text{actual}} = L / d$$

6000.0 / 454.0

-> **13.22**

5. Cracking Control

Estimated steel stress:

$$\sigma_s = 310 * (f_{yk}/500) * (A_{s,req}/A_{s,prov})$$

$$\sigma_s = 310 * (500.0/500) * (647.2/603.2)$$

-> **332.6 MPa**

Max allowable spacing:

From Table 7.3N for $w_k = 0.3$ mm

$\sigma_s = 332.6$ MPa -> interpolate

-> **84.2 mm**

Actual bar spacing:

$$s = (b - 2c - 2d_{\text{link}} - n*d_{\text{bar}}) / (n-1) + d_{\text{bar}}$$

$n = 3$, $d = 16.0$ mm

-> **104.0 mm**